Final Adventure

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Rules of Engagement

You may discuss the problems below with anyone in the following set of people:

$$P = \{\text{Anyone currently enrolled in MATH 3310, Kaylee Bodily, Caroline Torman, Erin Pitts, Brent Thomas}\}.$$

The documentation of your adventure should be created by teams of at most four and submitted by Sunday April 26, 2026 as a PDF via Canvas (the way we have done all semester). One submission per team.

No outside resources are to be used on this experience. Try to get through this with your brain and the brains of your team (if you choose to be part of one).

1 More Fun and Games

Funky Dice

Sub-Adventure 1.1

A six-sided die labeled with the integers $1, 2, 3, 4, 5, 6$ will be called a *standard die*. The goal for this part of the *Final* Adventure is to determine all ways to label a pair of dice with positive integers so that the probabilities of rolling the usual sums $2, 3, \ldots, 12$ are the same, but the labels are nonstandard.

(a) Let $p(x) = x + x^2 + x^3 + x^4 + x^5 + x^6$, and explain why $(p(x))^2$ is the generating function for the probabilities of outcomes in rolling a pair of standard dice.

(b) Let $A = (a_1, a_2, a_3, a_4, a_5, a_6)$ and $B = (b_1, b_2, b_3, b_4, b_5, b_6)$ be two lists of positive integers. Put $p_A(x) = x^{a_1} + x^{a_2} + x^{a_3} + x^{a_4} + x^{a_5} + x^{a_6}$ and $p_B(x) = x^{b_1} + x^{b_2} + x^{b_3} + x^{b_4} + x^{b_5} + x^{b_6}$. Explain why finding the a_i's and b_i's such that $p_A(x)p_B(x) = (p(x))^2$ is relevant to this part of the adventure.

(c) Factor $p(x)$ into irreducible polynomials and use this factorization to help solve for the a_i's and b_i's. Specifically, the factorization will force the form of $p_A(x)$ to be something like $p_1(x)^q p_2(x)^r p_3(x)^s p_4(x)^t$, where $0 \le q, r, s, t \le 2$ and $p_i(x)$, for $1 \le i \le 4$, is a factor of $p(x)$. In your solution to this step, you must motivate why you take this step.

(d) Begin to reduce the possibilities for q, r, s, and t by using information from $p_A(1)$ and $p_A(0)$. Note that, on one hand $p_A(1) = 1^{a_1} + 1^{a_2} + 1^{a_3} + 1^{a_4} + 1^{a_5} + 1^{a_6} = 6$ (since $a_i > 0$), and on the other hand we have $p_A(1) = p_1(1)^q p_2(1)^r p_3(1)^s p_4(1)^t$. Similarly, there are two ways to view $p_A(0)$.

(e) List all possible ways to label a pair of dice so that the probabilities of obtaining the sums $2, 3, 4, 5, 6, 7, 8, 9,\ 10, 11, 12$ are $\frac{1}{36}, \frac{2}{36}, \frac{3}{36}, \frac{4}{36}, \frac{5}{36}, \frac{6}{36}, \frac{5}{36}, \frac{4}{36}, \frac{3}{36}, \frac{2}{36}, \frac{1}{36}$, respectively. One such way will be the standard way. In your solution for this step, explain why you have proved that the labels you have found are the only possible ones that give the desired probabilities for the roll outcomes.

Solution:

Part (a):

$(p(x))^2$ is the generating function for the probabilities of outcomes in rolling a pair of standard dice because our generating function $p(x)$ is the possible outcomes of rolling a single die.

Each side of the die has the exact same chance of landing, so the constants are all 1.

When we roll two dice, we are taking the product of the generating functions for each die, (Shown through proposition 6. from our notes) which simply gives us $(p(x))^2$. This equation gives us the possibilities for rolling a pair of dice

Part (b):

We want to find the a_i's and b_i's such that $p_A(x)p_B(x) = (p(x))^2$ because we want to find the labels on the dice that give us the same probabilities as rolling a pair of normal dice.

Part (c):

First to explain, we needed to take this step so that we could find all the possible factors of $p_A(x)$ and $p_B(x)$. Since we know that $p_A(x)p_B(x) = (p(x))^2$, we can factor $p(x)$ into irreducible polynomials to find the possible factors of $p_A(x)$ and $p_B(x)$.

Actually doing it:

$$\begin{aligned} p(x) &= x + x^2 + x^3 + x^4 + x^5 + x^6 \\ &= x(1+x)(1+x^2)(1-x+x^2) \end{aligned}$$

We can see that the factors of $p(x)$ are $p_1(x) = x$, $p_2(x) = 1+x$, $p_3(x) = 1+x^2$, and $p_4(x) = 1-x+x^2$. Since we know that $p_A(x)p_B(x) = (p(x))^2$,

We can conclude that $p_A(x)$ must be of the form $p_1(x)^q p_2(x)^r p_3(x)^s p_4(x)^t$, where $0 \leq q, r, s, t \leq 2$.

We can also conclude that $p_B(x)$ must be of the form $p_1(x)^{2-q} p_2(x)^{2-r} p_3(x)^{2-s} p_4(x)^{2-t}$, where $0 \leq q, r, s, t \leq 2$.

Part (d):

Proof

From the power series expansion of e^x provided, let $x = -1$:

$$e^{-1} = \frac{1}{e} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} + \frac{(-1)^{n+1}}{(n+1)!} + \ldots$$

Multiplying this by $n!$ gives:

$$\frac{n!}{e} = n! \left(\sum_{k=0}^{n} \frac{(-1)^k}{k!} + \sum_{k=n+1}^{\infty} \frac{(-1)^k}{k!} \right)$$

Substituting the formula for D_n derived in Part (c), we have:

$$\frac{n!}{e} = D_n + n! \sum_{k=n+1}^{\infty} \frac{(-1)^k}{k!}$$

$$D_n - \frac{n!}{e} = -n! \left(\frac{(-1)^{n+1}}{(n+1)!} + \frac{(-1)^{n+2}}{(n+2)!} + \dots \right)$$

The term $E = \left| D_n - \frac{n!}{e} \right|$ is an alternating series. Based on our notes by the properties of alternating series, the remainder is less than the first omitted term:

$$E < \left| \frac{n!}{(n+1)!} \right| = \frac{1}{n+1}$$

For $n \geq 1$, the term $\frac{1}{n+1} \leq \frac{1}{2}$. Since the distance between D_n and $\frac{n!}{e}$ is less than 1/2, D_n must be the nearest integer to $\frac{n!}{e}$.

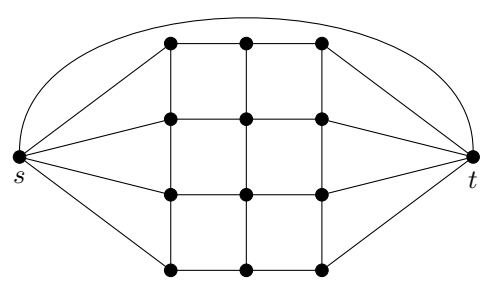
Conclusion:

This proves that $D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor$.

Rocky vs Bullwinkle

Sub-Adventure 1.2

Two players Rocky and Bullwinkle alternate coloring edges in the graph below. Rocky goes first and colors an edge red. Bullwinkle goes next and colors an edge blue. The game is over when Bullwinkle has colored a blue path that connects the vertex s to the vertex t or when all the edges have been colored. Bullwinkle wins if there is a blue path from s to t, otherwise Rocky wins.

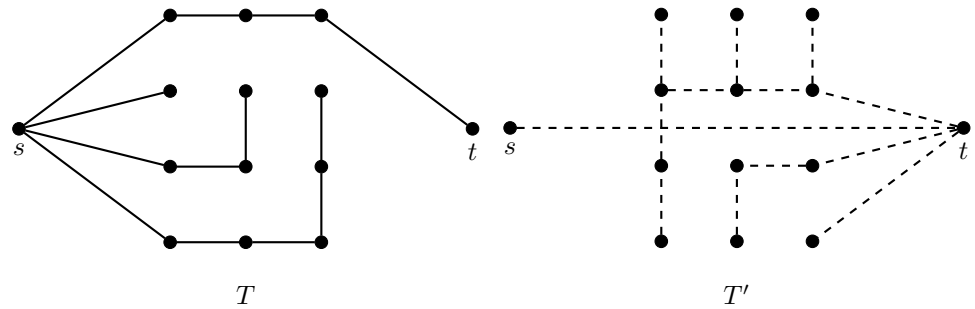


Confirm or deny, with proof, whether Bullwinkle can always win if he employs a particular strategy for each move.

I suggest the following approach.

- Show the graph has two disjoint spanning trees (spanning trees that share no edges).
- Find an equivalent game that involves one player removing an edge of the graph and the other player drawing a multiedge instead of coloring edges. In other words, one player 'colors' the edge by erasing it and the other player 'colors' an edge by doubling it.
- Use theorems we have proved or will prove in class to argue your assertion.

Solution: Bullwinkle can always win If he uses the correct strategy.



As shown above the graph can be split into two disjoint spanning trees. We will call the two trees T and T'. Both trees span all vertices without sharing any edges.

It is better for our purpose to create an equivalent game. Rocky removes single edges instead of coloring them red and Bullwinkle doubles an edge making it impossible for rocky to remove it. Bullwinkle wins the game if there is a path of doubled edges from S to T.

The strategy for Bullwinkle is to use theorem 40 in the notes. According to the theorem "if T and T' are spanning trees of a connected graph G and $e \in E(T) - E(T')$, then there is an edge $e' \in E(T') - E(T)$ such that $T - e + e'$ is a spanning tree of G." This means that when rocky removes an edge e from tree T there will always be an edge e' from T' that will reconnect that tree.

In practice the strategy is to pay attention to when rocky disconnects the spanning tree and to make sure to reconnect it with an edge from the second spanning tree. This guarantees that there will always be a path from s to t through the spanning tree.

2 Invitation Madness

Sub-Adventure 2

You decide to use snail mail to send out invitations to a game night. Since you have never sent a piece of paper mail before in your life you start to worry that no one will receive the correct invitation. It occurs to you that there are more ways for everyone you invited to receive an invitation addressed to someone else than there are for everyone to receive the correct invitation. This only serves to exacerbate your anxiety. You decide to calculate the number of ways that all of the invitations can be sent to the wrong person.

Determine the number D_n of ways the n invitations can be delivered to your n friends so that no one gets the invitation actually intended for them.

Moreover, it is asked that you determine D_n in a few ways: one way entails finding a recurrence relation for D_n, another entails a closed (albeit complicated) formula for D_n, and finally a fairly simple formula for D_n obtained from the more complicated one.

(a) Prove the recurrence $D_n = (n-1)D_{n-1} + (n-1)D_{n-2}$, for $n \geq 2$, with $D_0 = 1$ and $D_1 = 0$,

(b) Derive, from the recurrence in sub-adventure 2(a), this recurrence $D_n = nD_{n-1} + (-1)^n$, for $n \geq 1$, with $D_0 = 1$.

(c) Use the Principle of Inclusion-Exclusion to compute a closed formula for D_n.

(d) Noting that e^x has a power series expansion

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots ,$$

prove that D_n is the nearest integer to $\frac{n!}{e}$; that is, show that

$$D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

(Now that's a nice and concise closed formula!)

Solution:

Part (a):

Proof

We can prove the recurrence $D_n = (n-1)D_{n-1} + (n-1)D_{n-2}$ by using a combitorial argument. First, we can choose any of the $n-1$ wrong invitations to send to the first person. Then, we have two cases.

In the first case, we can send the invitation intended for the person who received the first person's invitation to the first person. In this case, there are D_{n-2} ways to count the remaining $n-2$ invitations.

In the second case, we can send the invitation intended for the person who received the first person's invitation to someone else. In this case, there are D_{n-1} ways to count the remaining $n-1$ invitations.

Since there are $n-1$ ways to choose which wrong invitation to send to the first person and since these cases are mutually exclusive, we have that $D_n = (n-1)D_{n-1} + (n-1)D_{n-2}$.

Conclusion:

Thus, we have proved the recurrence $D_n = (n-1)D_{n-1} + (n-1)D_{n-2}$, for $n \geq 2$, with $D_0 = 1$ and $D_1 = 0$.

Part (b):

Proof

We can derive the recurrence $D_n = nD_{n-1} + (-1)^n$ from the recurrence in sub-adventure 2(a) by using induction.

Base Case:

When $n = 1$, we have $D_1 = 0$ and $1D_0 + (-1)^1 = 1(1) + (-1) = 0$. Thus, the base case holds.

Induction Step:

Assume that $D_k = kD_{k-1} + (-1)^k$ for all $k < n$. We want to show that $D_n = nD_{n-1} + (-1)^n$. By the recurrence in sub-adventure 2(a), we know that $D_n = (n-1)D_{n-1} + (n-1)D_{n-2}$. By the induction hypothesis, we have that $D_{n-1} = (n-1)D_{n-2} + (-1)^{n-1}$ and $D_{n-2} = (n-2)D_{n-3} + (-1)^{n-2}$.

After doing math, we can see that:

$$\begin{aligned}
D_n &= (n-1)D_{n-1} + (n-1)D_{n-2} \\
&= (n-1)((n-1)D_{n-2} + (-1)^{n-1}) + (n-1)((n-2)D_{n-3} + (-1)^{n-2}) \\
&= (n-1)(n-1)D_{n-2} + (n-1)(-1)^{n-1} + (n-1)(n-2)D_{n-3} + (n-1)(-1)^{n-2} \\
&= (n-1)(n-1)D_{n-2} + (n-1)(n-2)D_{n-3} + (n-1)(-1)^{n-1} + (n-1)(-1)^{n-2} \\
&= (n-1)(n-1)D_{n-2} + (n-1)(n-2)D_{n-3} + (n-1)((-1)^{n-1} + (-1)^{n-2}) \\
&= (n-1)(n-1)D_{n-2} + (n-1)(n-2)D_{n-3} + (n-1)(-1)^{n-2}((-1)+1) \\
&= (n-1)(n-1)D_{n-2} + (n-1)(n-2)D_{n-3} \\
&= (n-1)((n-1)D_{n-2} + (n-2)D_{n-3}) \\
&= (n-1)D_{n-1} \\
&= nD_{n-1} + (-1)^n
\end{aligned}$$

Conclusion:

Thus, we have shown that $D_n = nD_{n-1} + (-1)^n$.

Part (c):

To compute a closed formula for D_n using the Principle of Inclusion-Exclusion, let S be the set of all $n!$ possible permutations of delivering n invitations to n friends. Let A_i be the set of deliveries where friend i receives their correct invitation. We are looking for D_n, the number of deliveries where no one receives the correct invitation, which is $|\bigcap_{i=1}^{n} \overline{A_i}|$.
According to the Principle of Inclusion-Exclusion:

$$D_n = |S| - \sum |A_i| + \sum |A_i \cap A_j| - \cdots + (-1)^n |A_1 \cap \cdots \cap A_n|$$

For k $i_1, i_2, \ldots, i_k$, the size of the intersection $|A_{i_1} \cap \cdots \cap A_{i_k}|$ is the number of ways to arrange the invitations such that those k specific people get the right ones, while the remaining $n-k$ are distributed in any of $(n-k)!$ ways. Since there are $\binom{n}{k}$ ways to choose these k people, we have:

$$\begin{aligned}
D_n &= \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! \\
&= \sum_{k=0}^{n} (-1)^k \frac{n!}{k!(n-k)!} (n-k)! \\
&= \sum_{k=0}^{n} \frac{(-1)^k n!}{k!} \\
&= n! \left(1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right)
\end{aligned}$$

Conclusion:

The closed formula is $D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$.

Part (d):

Proof

As we figured out in part C we get write our formula as:

$$\frac{1}{e} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = \left(\sum_{k=0}^{n} \frac{(-1)^k}{k!} \right) + \left(\sum_{k=n+1}^{\infty} \frac{(-1)^k}{k!} \right)$$

Multiplying through by $n!$, we get:

$$\frac{n!}{e} = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} + n! \sum_{k=n+1}^{\infty} \frac{(-1)^k}{k!}$$

If we substitute our formula for D_n from Part (c):

$$\frac{n!}{e} = D_n + \left(\frac{(-1)^{n+1}}{n+1} + \frac{(-1)^{n+2}}{(n+1)(n+2)} + \dots \right)$$

Let the error term be $R_n = \frac{n!}{e} - D_n$. This is an alternating series where the absolute value of the terms is strictly decreasing. By the Alternating Series Estimation Theorem, the magnitude of the remainder is less than the first omitted term:

$$|R_n| = \left| \frac{n!}{e} - D_n \right| < \frac{1}{n+1}$$

For $n \geq 2$, we have $|R_n| < \frac{1}{3} < \frac{1}{2}$. (For $n = 1$, $D_1 = 0$ and $|0 - 1/e| \approx 0.36 < 0.5$). Since the distance between the integer D_n and the value $n!/e$ is strictly less than $1/2$, D_n must be the integer closest to $n!/e$.

Conclusion:

Therefore, $D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor$.

3 An Optimization Problem

Sub-Adventure 3

Let H denote an arbitrary graph. The distance between vertices in H is the length of the shortest path that has those vertices as its endpoints. Denote by $d_{\max}(H)$ the maximum distance among all pairs of vertices of H. Note that $\Delta(H)$ denotes the maximum degree among all vertices of H.

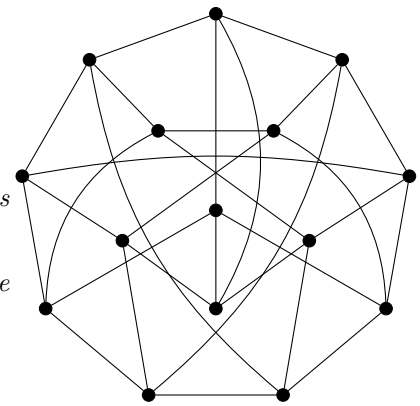
Define the function $N(d, k)$ to be the maximum number of vertices among all graphs H with $d_{\max}(H) = d$ and $\Delta(H) = k$.

(a) *Determine* $N(n, 2)$.

(b) *Determine* $N(2, 3)$.

(c) *Verify that the graph* G *drawn to the right has* $d_{\max}(G) = 2$.

(d) Give bounds for $N(2, 4)$. *Bonus: Determine* $N(2, 4)$.



Solution: For the following parts, we'll imagine a fantasy land of wizards. A graph will be referred to as a guild, as in a guild of wizards, a vertice will be a wizard tower and the wizards use connection spells to communicate, these are our edges.

Part A

Wizards are starting to work together to form a chain of communication spells...

$\Delta H = 2$, every tower is connected to atmost 2 others. $d_{max}H = n$, Furthest any two towers can be is n steps.

If every tower has at most 2 connections, the graph can only be a path or a cycle.

- In a cycle, the furthest apart any two vertices can be is half way around the cycle.
- In a path, the furthest distance is between the two ends.

Taking the longest distance of a cycle as just a path, we know that there are $n + 1$ vertices, and n edges.

Therefore, $N(n, 2) = n + 1$.

Part B

A guild of wizards are now banding together, they want to have fast communication between all of them, unfortunately, they take a lot of power so one wizard tower can only support 3 connection spells. The guild wants to be well connected so they also only want to be two connection spells away at max.

Let's take a tower, T, as a one item set containing any one tower of the guild. The set T_2 is made of the neighboring towers of T. There can be 3 connections to T, so there are 3 neighbors.

The set T_3 is the far neighbors of T, they connect to the towers of T_2. Since the towers of T_2 use one connection spell connecting to T they can use two new connections for new towers. T_3 will have two towers for every tower in T_2.

Since the guild doesn't want the distance to be more than 2, we cannot have far-far neighbors, we stop here.

1. T has one tower
2. T_2 has three towers
3. T_3 has six towers (3 towers $\times 2$ spells)

Total = 10 towers. Since this works for any one tower in the graph, we see that the max is 10. $N(2,3) = 10$

Part C

Ignoring the wizards

The graph has 15 vertices and four edges each. In the classnotes, <u>Theorem 33</u> states..

Theorem 33. Suppose G is a graph and $V(G) = \{v_1, v_2, ..., v_n\}$. Then if A is the adjacency matrix of G, entry (i, j) in A^k is the number of walks of length k from v_i to v_j.

We can take the first and second order adjacency matrix and compare. If all entries in $A^1 + A^2$ are greater than 0, the max distance is 2.

$$A = \begin{bmatrix} 0&1&0&0&0&0&0&0&1&0&0&0&0&1&1 \\ 1&0&1&0&0&1&0&0&0&0&0&1&0&0&0 \\ 0&1&0&1&0&0&0&1&0&0&0&0&1&0&0 \\ 0&0&1&0&1&0&0&0&0&0&0&1&0&1&0 \\ 0&0&0&1&0&1&0&0&1&0&0&0&1&0&0 \\ 0&1&0&0&1&0&1&0&0&1&0&0&0&0&0 \\ 0&0&0&0&0&1&0&1&0&0&1&0&0&1&0 \\ 0&0&1&0&0&0&1&0&1&1&0&0&0&0&0 \\ 1&0&0&0&1&0&0&1&0&0&1&0&0&0&0 \\ 0&0&0&0&0&1&0&1&0&0&0&1&0&0&1 \\ 0&0&0&0&0&0&1&0&1&0&0&1&1&0&0 \\ 0&1&0&1&0&0&0&0&0&1&1&0&0&0&0 \\ 0&0&1&0&1&0&0&0&0&0&1&0&0&0&1 \\ 1&0&0&1&0&0&1&0&0&0&0&0&0&0&1 \\ 1&0&0&0&0&0&0&0&0&1&0&0&1&1&0 \end{bmatrix} \quad A^2 = \begin{bmatrix} 4&0&1&1&1&1&1&1&0&1&1&1&1&1&1 \\ 0&4&0&2&1&0&1&1&1&2&1&0&1&1&1 \\ 1&0&4&0&2&1&1&0&1&1&1&2&0&1&1 \\ 1&2&0&4&0&1&1&1&1&1&1&0&2&0&1 \\ 1&1&2&0&4&0&1&1&0&1&2&1&0&1&1 \\ 1&0&1&1&0&4&0&2&1&0&1&2&1&1&1 \\ 1&1&1&1&1&0&4&0&2&2&0&1&1&0&1 \\ 1&1&0&1&1&2&0&4&0&0&2&1&1&1&1 \\ 0&1&1&1&0&1&2&0&4&1&0&1&2&1&1 \\ 1&2&1&1&1&0&2&0&1&4&1&0&1&1&0 \\ 1&1&1&1&2&1&0&2&0&1&4&0&0&1&1 \\ 1&0&2&0&1&2&1&1&1&0&0&4&1&1&1 \\ 1&1&0&2&0&1&1&1&2&1&0&1&4&1&0 \\ 1&1&1&0&1&1&0&1&1&1&1&1&1&4&1 \\ 1&1&1&1&1&1&1&1&1&0&1&1&0&1&4 \end{bmatrix}$$

$$A + A^2 = \begin{bmatrix} 4&1&1&1&1&1&1&1&1&1&1&1&1&2&2 \\ 1&4&1&2&1&1&1&1&1&2&1&1&1&1&1 \\ 1&1&4&1&2&1&1&1&1&1&1&2&1&1&1 \\ 1&2&1&4&1&1&1&1&1&1&1&1&2&1&1 \\ 1&1&2&1&4&1&1&1&1&1&2&1&1&1&1 \\ 1&1&1&1&1&4&1&2&1&1&1&2&1&1&1 \\ 1&1&1&1&1&1&4&1&2&2&1&1&1&1&1 \\ 1&1&1&1&1&2&1&4&1&1&2&1&1&1&1 \\ 1&1&1&1&1&1&2&1&4&1&1&1&2&1&1 \\ 1&2&1&1&1&1&2&1&1&4&1&1&1&1&1 \\ 1&1&1&1&2&1&1&2&1&1&4&1&1&1&1 \\ 1&1&2&1&1&2&1&1&1&1&1&4&1&1&1 \\ 1&1&1&2&1&1&1&1&2&1&1&1&4&1&1 \\ 2&1&1&1&1&1&1&1&1&1&1&1&1&4&2 \\ 2&1&1&1&1&1&1&1&1&1&1&1&1&2&4 \end{bmatrix}$$

There are no 0 entries in the $A+A^2$ matrix, therefore according to the theorem, the distance between any two points is at most 2.

Part D

The wizards have researched their connection spells and have made them more efficient, a tower can now sustain 4 connection spells at once.

Theoretically, to find the max amount of towers the guild can cram in, we assume the guild branches out as efficiently as possible.

1. T has one tower (and can have 4 neighbors)
2. T_2 has four towers
3. T_3 has 3 towers for each tower in T_2, 12 in total.

The total amount of towers in this theoretical guild is 17 towers. The upper bound is 17 towers

For the lower bound, it's already been proven that 15 towers is possible, going lower does not help as we're looking for the max.

Let's hope these wizards aren't up to something evil...

4 Stirling Numbers

Sub-Adventure 4.1

Let $\mathcal{N}$ denote the set of all possible ways to place s labeled balls into t labeled boxes so that no box is empty. Recall from your midterm that $|\mathcal{N}| = t!\left\{ {s \atop t} \right\}$. Use the recurrence $\left\{ {n \atop k} \right\} = k\left\{ {n-1 \atop k} \right\} + \left\{ {n-1 \atop k-1} \right\}$ to create a generating function for the Stirling numbers by following the steps below.

(a) Let k be a fixed but arbitrary number and set

$$f_k(x) = \sum_{n=k}^{\infty} \left\{ {n \atop k} \right\} x^n$$

Explain why the sum does not start when $n = 0$ and show $kxf_k(x) + xf_{k-1}(x) = f_k(x)$.

(b) Solve for $f_k(x)$ in the equation from part (a) and then iterate the resulting recurrence to obtain

$$f_k(x) = x^k \prod_{r=1}^{k} \frac{1}{(1 - rx)}$$

(c) Now calcluate $|\mathcal{N}|$ by using the Principle of Inclusion–Exclusion. Use your result to conclude that

$$\left\{ {s \atop t} \right\} = \sum_{k=0}^{t} \frac{(-1)^k (t-k)^s}{k!(t-k)!}$$

Solution:

Part (a):

The sum does not start when $n = 0$ because $\left\{ {n \atop k} \right\}$ is defined to be 0 when $n < k$. Because we can't place more balls than boxes, so sum starts at $n = k$. Thus, the first nonzero term in the sum occurs when $n = k$. It also is so that we can use the recurrence relation for the stirling numbers of the second kind.

To show that $kxf_k(x)+xf_{k-1}(x) = f_k(x)$, we can use the recurrence relation for the Stirling numbers of the second kind as shown in our notes. We have:

$$kxf_k(x) + xf_{k-1}(x) = \sum_{n=k+1}^{\infty} k\left\{ {n-1 \atop k} \right\} x^n + \sum_{n=k}^{\infty} \left\{ {n-1 \atop k-1} \right\} x^n \quad = \sum_{n=k+1}^{\infty} (k\left\{ {n-1 \atop k} \right\} + \left\{ {n-1 \atop k-1} \right\})x^n + \left\{ {\mathbf{k-1} \atop \mathbf{k-1}} \right\} \mathbf{x^k} \quad = \sum_{n=}^{\infty}$$

So we have shown that $kxf_k(x) + xf_{k-1}(x) = f_k(x)$.

Part (b):

To solve for $f_k(x)$ in the equation from part (a), we will rearrange the equation to get:

$$\begin{aligned} f_k(x) - kxf_k(x) &= xf_{k-1}(x) \\ f_k(x)(1-kx) &= xf_{k-1}(x) \\ f_k(x) &= \frac{xf_{k-1}(x)}{1-kx} \end{aligned}$$

Now we can iterate this recurrence to obtain:

$$\begin{aligned} f_k(x) &= \frac{xf_{k-1}(x)}{1-kx} \\ &= \frac{x}{1-kx} \cdot \frac{xf_{k-2}(x)}{1-(k-1)x} \\ &= \frac{x^2 f_{k-2}(x)}{(1-kx)(1-(k-1)x)} \\ &= \frac{x^2}{(1-kx)(1-(k-1)x)} \cdot \frac{xf_{k-3}(x)}{1-(k-2)x} \\ &= \frac{x^3 f_{k-3}(x)}{(1-kx)(1-(k-1)x)(1-(k-2)x)} \\ &= \dots \\ &= \frac{x^k f_0(x)}{\prod_{r=1}^{k}(1-rx)} \end{aligned}$$

We know that $f_0(x) = \sum_{n=0}^{\infty} \left\{ {n \atop 0} \right\} x^n$. Since $\left\{ {n \atop 0} \right\} = 0$ for all $n > 0$ and $\left\{ {0 \atop 0} \right\} = 1$, we have $f_0(x) = 1$. Thus, we can simplify to get:

$$f_k(x) = \frac{x^k}{\prod_{r=1}^{k}(1-rx)}$$

Conclusion:

Thus, we have shown that $f_k(x) = x^k \prod_{r=1}^{k} \frac{1}{(1-rx)}$.

Part (c):

To calculate $|\mathcal{N}|$ by using the Principle of Inclusion-Exclusion, we can let A_i be a set where box i is empty. We want to find the number of arrangements where no box is empty, which is $|\bigcap_{i=1}^{t} \overline{A_i}|$.

By the Principle of Inclusion-Exclusion as included in our notes, we have:

$$\begin{aligned}|\bigcap_{i=1}^{t}\overline{A_i}| &= \sum_{k=0}^{t}(-1)^k\binom{t}{k}(t-k)^s \\ &= t!\sum_{k=0}^{t}\frac{(-1)^k(t-k)^s}{k!(t-k)!}\end{aligned}$$

Since $|\mathcal{N}| = t!\left\{{s \atop t}\right\}$, we can conclude that $\left\{{s \atop t}\right\} = \sum_{k=0}^{t}\frac{(-1)^k(t-k)^s}{k!(t-k)!}$.

Conclusion:

Thus, we have shown that $\left\{{s \atop t}\right\} = \sum_{k=0}^{t}\frac{(-1)^k(t-k)^s}{k!(t-k)!}$.

5 Flocks

Queens

Sub-Adventure 5.1

Queens. Let n and q be positive integers with $q \leq n$. Define an (n, q)-flock to be a flock on n vertices with exactly q queens. In this sub-adventure, you will determine all possible values of n and q for which there exists an (n, q)-flock. Specifically, please prove the following theorem.

Theorem 1. *There exists an (n, q)-flock for all positive integers n and q with $n \geq q \geq 1$, except for $q = 2$ and arbitrary n, and $n = q = 4$.*

I suggest that you use the Principle of Mathematical Induction and the following lemmas (which I ask that you prove as well).

Lemma 1. *There exists a $(1, 1)$-flock.*

Lemma 2. *There does not exist a $(4, 4)$-flock.*

Lemma 3. *There is a $(6, 6)$-flock.*

Lemma 4. *If there exists an (k, k)-flock, then there exists an $(k + 2, k + 2)$-flock, where k is a positive integer.*

Lemma 5. *For every positive integer n except 2 and 4, there exists an (n, n)-flock.*

Solution:

Proof of Lemma 1: There exists a $(1, 1)$-flock.

As there is only one queen and one vertex, there are no restrictions on the flocks, thus satisfying the distance requirement.

Proof of Lemma 2: There does not exist a $(4, 4)$-flock.

When $n = q = 4$, the tournament is too constrained for every chicken to reach every other chicken in 1 or 2 steps.

We can prove this by method of exhaustion. If we assume a chicken A pecks only one other chicken, it would need that chicken to peck all others to reach them in 2 steps. However, this orientation prevents the other chickens from reaching A within the 2-step limit. After testing all possible edge orientations for 4 chickens, there is no configuration where all four chickens qualify as queens.

Proof of Lemma 3: There is a $(6, 6)$-flock.

We can prove this by providing a valid example. If we arrange 6 chickens and orient the edges such that each chicken pecks the two chickens immediately following it in a circle, and also pecks one chicken across the circle, every chicken can reach any other in at most 2 steps.

This solution has all chickens acting as queens.

Proof of Lemma 4: If there exists an (k, k)-flock, then there exists an $(k + 2, k + 2)$-flock, where k is a positive integer.

We can prove this by method of induction, for both even and odd numbers.

For odd numbers:

Base case: Using lemma 1, we know that there exists a $(1, 1)$-flock.

Induction Hypothesis: This relationship holds true for some $k \geq 1$ (k being an odd positive integer), that is, there exists a (k, k)-flock.

Induction Step: We must now show that this relationship continues to hold true for $k + 2$, that is, there exists a $(k + 2, k + 2)$-flock.

We can do this by introducing 2 new chickens, A and B. We can then arrange them in a manner such that A pecks B, B pecks every chicken in the original flock, and every chicken in the original flock pecks A. Thus, the relationship holds true for $k + 2$.

Conclusion: The relationship is true for $n = q = 1$, and the relationship's truth for odd numbers of $n = q = k \geq 1$ implies its truth for $n = q = k + 2$ and so the relationship is true for all odd nonnegative integers by the Principle of Mathematical Induction.

For even numbers:

Base case: Using lemma 2, we know that there does not exist a $(4, 4)$-flock. Following that, we can treat the $(4, 4)$-flock as our $(k + 2, k + 2)$-flock, making our (k, k)-flock equivalent to the $(2, 2)$-flock. As we know that the $(4, 4)$-flock does not exist, it follows that the $(2, 2)$-flock also does not exist. However, by Lemma 3, we know that there exists a $(6, 6)$-flock, making that our base case.

Induction Hypothesis: This relationship holds true for some $k \geq 6$ (k being an even positive integer), that is, there exists a (k, k)-flock.

Induction Step: We must now show that this relationship continues to hold true for $k + 2$, that is, there exists a $(k + 2, k + 2)$-flock.

We can do this by introducing 2 new chickens, A and B. We can then arrange them in a manner such that A pecks B, B pecks every chicken in the original flock, and every chicken in the original flock pecks A. Thus, the relationship holds true for $k + 2$.

Conclusion: The relationship is true for $n = q = 6$, and the relationship's truth for odd numbers of $n = q = k \geq 6$ implies its truth for $n = q = k + 2$ and so the relationship is true for all even nonnegative integers by the Principle of Mathematical Induction.

Proof of Lemma 5: For every positive integer n except 2 and 4, there exists an (n, n)-flock.

Referring to the conclusions drawn from the proof of Lemma 4, we see that for every odd positive integer $n \geq 1$, there exists an (n, n)-flock. We also see that for every even positive integer $n \geq 6$, there exists an (n, n)-flock.

Thus, we know that for every positive integer n except 2 and 4, there exists an (n, n)-flock, confirming Lemma 5.

Theorem 1: There exists an (n, q)-flock for all positive integers n and q with $n \geq q \geq 1$, except for $q = 2$ and arbitrary n, and $n = q = 4$.

Proof of Theorem 1:

Base Case: Using Lemma 1, we know that there exists a $(1, 1)$-flock ($n = 1$). From Lemma 5, we know that there exists for every positive integer n an (n, n)-flock, except 2 and 4.

Induction Hypothesis: This relationship holds true for some positive integer k, that is, there exists a (k, q)-flock for all $1 \leq q \leq k$, except when $q = 2$ or when $k = q = 4$.

Induction Step: We must now show that this relationship continues to hold true for $k + 1$, that is, there exists a $(k + 1, k + 1)$-flock. This was done in the proof of Lemma 4.

We must also show that this relationship exists when $q \leq k$. To create a flock of $(k+1, q)$, we simply add another chicken in a manner every original queen pecks this new chicken without becoming a queen. This satisfies the requirement.

Conclusion: The relationship is true for $n = k$, and thus implies its truth for $n = k + 1$ and so the relationship is true for all nonnegative integers (except for 2 and 4) by the Principle of Mathematical Induction.